

LOC 8.0

LINES OF CODE

- **Problem Statement** - Smart College Hackathon Management & Verification System with QR-Based Access and Evaluation
- **Team Name** - SideQuest



Smart Hackathon Management System

A Unified, Fraud-Proof, AI-Verified Digital Ecosystem - Not Just Another Registration Tool



Biometric-Verified Digital Identity

Unlike Google Forms or manual checks, we use

- AI Face Match + OTP + Admin Approval
- Masked ID verification

→ Ensures zero fake or duplicate registrations



AI Judging Co-Pilot (Game Changer)

Judges don't just "watch demos" - they get AI assistance:

- Automatic PPT content extraction & AI scoring
- GitHub plagiarism detection
- Code Quality vs PPT claim comparison
- Auto score aggregation & live ranking

→ Makes judging transparent, structured & tamper-proof



Anti-Screenshot Dynamic QR Ecosystem

Not static QR. Not printable passes.

- 30-second rotating TOTP QR
- One-time cryptographic meal tokens
- <200ms offline scanning (works without Wi-Fi)

→ Eliminates queue chaos & coupon fraud completely



Sponsor ROI Heatmap & Full Audit Trail

- Real-time booth footfall tracking
- Timestamped logs of every entry, scan & score
- QR-verifiable Auto Certificate Generator

→ Delivers measurable impact, not "vibe-based" reporting



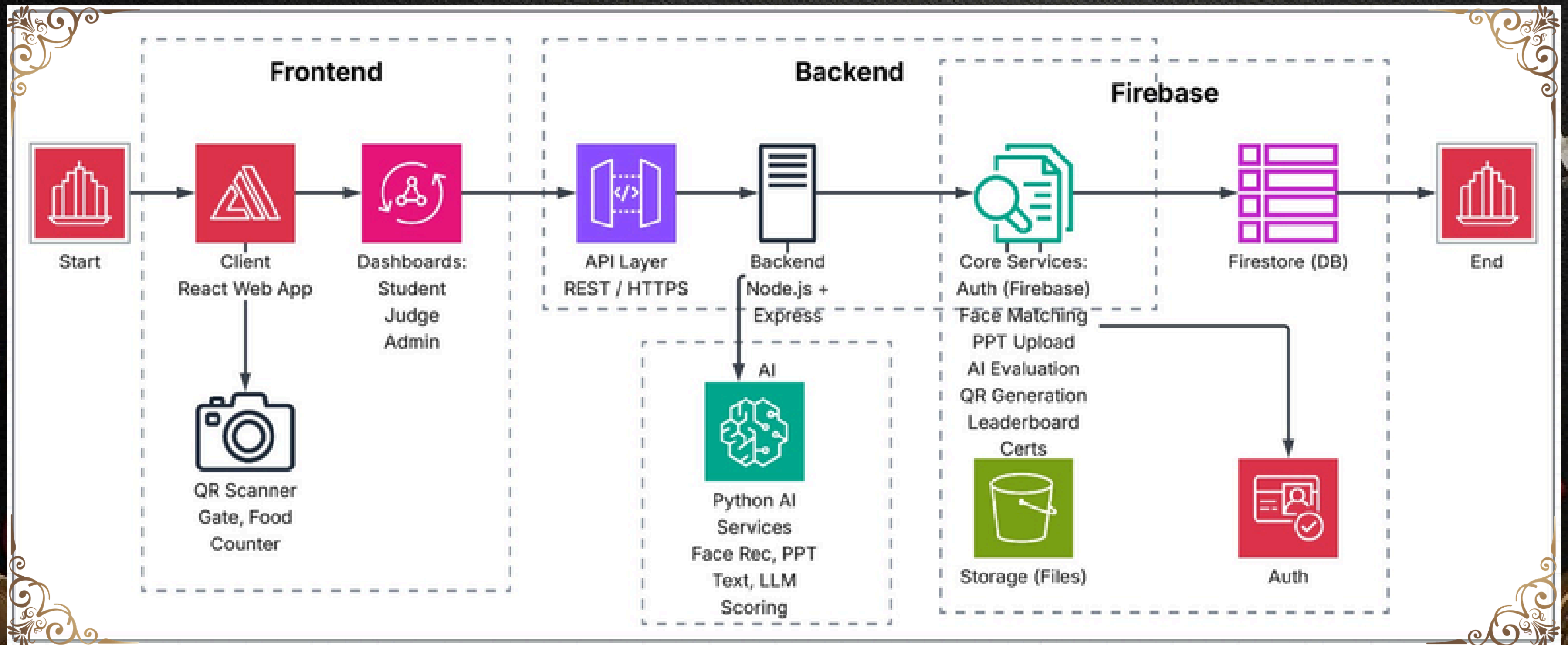
Why We're Unique



- Offline-First Mesh Sync (No internet dependency)
- Unified Entry + Food + Attendance + Judging
- Real-time Fraud Detection Flags
- Designed as a Scalable Campus Event OS

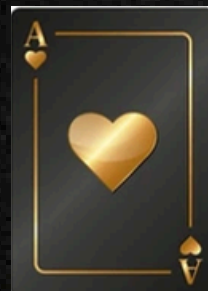


TECHNICAL APPROACH



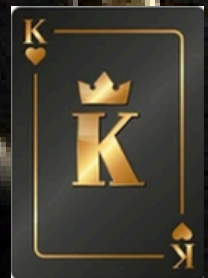
All-in on Resilience : Feasibility & Risk Strategy

Feasibility



Ace of Technology

- Proven Face APIs + TOTP QR
- Standard Scalable Tech Stack



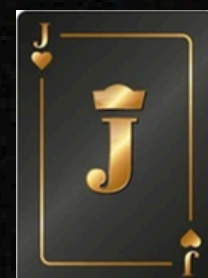
King of Efficiency

- No hardware needed
- Smartphone powered System



Queen of Cost Control

- QR validation handled locally in app
- Minimal server calls



Jack of Scalability

- Modular build
- Entry, Food, Judging
- Built and Tested easily

Beating the Risks



Network crash?

- Switch to Offline Mode.
- Scanner stores verified participants and syncs when internet is back.



QR Screenshot Sharing

- Use short-validity QR
- Refreshes every 30-60 sec
- Backend verifies timestamp before approval



Face Match Errors

- Confidence threshold + manual override
- Low confidence => Admin approval
- Not fully dependent on automation



Data Privacy

- Minimal Storage Policy
- Store only verified name+ID reference
- Delete ID images after verification
- Encrypt sensitive fields

Resilience First



Our Royal flush

- Works when Wifi folds
- Prevents fraud before it plays
- Designed for crowd chaos
- 100 Registrations = 100 verified meals

OTHERS PLAY THE GAME.

WE CONTROL THE TABLE.

POTENTIAL IMPACT & REAL-WORLD BENEFITS

IMPACT ON TARGET AUDIENCE

For Participants

- Faster Entry => No long queues
- Fair judging => Reduced bias and plagiarism
- Transparent Results => Trust in rankings
- No physical coupons => Everything digital

For Organisers

- Real time monitoring dashboard
- Instant attendance & food tracking
- Structured Judging system

For Sponsors

- No physical coupons and food tracking
- Clear event reports for ROI justifications

BROADER BENEFITS

Economis Benefits

- Reduces printing costs
- Minimizes food wastage
- Deployable across multiple campus events

Environmental Benefits

- Paperless Registrations
- No printed meal coupons
- Reliable participation metrics
- Digital audit trail instead of physical logs

Social Benefits

- Reduces conflict over results
- Encourages accountability in campus events
- Builds trust between students and administration

THE BIGGER PICTURE

- Makes Hackathon Fair and transparent
- Makes organising more efficient
- Makes campuses events digitally accessible
- Makes sponsors digital ready

**WHEN THE SYSTEM
IS FAIR AND
EFFICIENT**

**EVERYONE WINS AT
THE TABLE**